**TABLE 2**

**IoT PLATFORMS FOCUSED ON DEVICE MANAGEMENT AND AI**

| Details | C3 AI | Particle IoT Platform | Kaa IoT Platform |
|---|---|---|---|
| Type | SaaS solution | PaaS solution | PaaS solution |
| Integrations | Its architecture is scalable with cloud services like AWS Cloud | It can integrate with GCP, Azure IoT Hub, IFTTT, QuestDB | REST API, WebSockets, Data Analytics, Business tools like SAP, Salesforce |
| API Support | REST APIs, Customisable API integration | REST API | REST API |
| Features | • Data Query & Analysis<br>• Data Integration<br>• Data Abstraction<br>• Machine Learning<br>• Application Management<br>• Application Logic<br>• Application Deployment<br>• Operations Workflow Management | • Edge-to-Cloud Stack<br>• IoT Device Management<br>• Data Pipeline<br>• IoT Hardware and Tracking | • Handle millions of devices<br>• Ease of use<br>• Third-party integration<br>• Data security<br>• Multicloud architecture |
| Noteworthy Character-istics | Users can select from a set of pre-built applications. The C3.ai platform also provides a development environment for building custom IoT applications | Particle supports Over-The-Air (OTA) firm-ware updates. Also, the program structure for Particle is built with Arduino compatibility. | It is a completely open source solution |
| Interface | Web based, Desktop | Web IDE, Desktop IDE(Dev) and a CLI (Com-mand Line Interface) | N/A |
| Visualisation | Reports, dashboards, visualisations, and actionable insights | Dashboard, Google Maps | Built-in Dashboard or third-party tools like Grafana. Users can stream data from IoT devices to any data analytics system via the pre-integrated Kafka channel. |
| Pricing | Depends on features selected, Quote-based | Three pricing levels: Enterprise, Starter, Tracking | Multiple plans are available, depending on hosting and number of devices |
| Website | https://c3.ai/c3-ai-suite/ | https://www.particle.io/what-is-particle/ | https://www.kaaiot.com/ |

during the implementation phase, which makes it easy for developers to get started.

## 7. Disaster recovery and downtime

In any application, it is only natural for technical problems to occur. What matters is how the parent company deals with such problems, and how long its recovery takes. While selecting an IoT platform for your application, you need to make sure that the platform provides the necessary infrastructure to handle such occurrences without any data loss. Check if the company provides a backup solution for your data, either locally or on cloud.

## 8. Updates and upgrades

Although it is mainstream, IoT can still be called a relatively new technology. Your needs might change, your company might grow. So, does your IoT platform support such updates in your company's infrastructure? Does it support the newer versions of IoT protocols? Does it provide facilities to update your firmware over-the-air? If yes, are these facilities in your budget? These are important questions that you need to ask before taking any decision and selecting an IoT platform.

## 9. Device management

IoT device management is significantly crucial because all your devices need to be secure and connected for your system to function properly. Ideally, the IoT platform should enable you to register a device easily, provide means to delete or update it, provide its status updates, and notify you if it gets disconnected from the system or malfunctions. All of this needs to be done in real time.

Moreover, the platform should enable you to remotely control your devices over the internet. It should support a wide variety of devices and should have a good mechanism to sort them into types. It should also have low latency even if the number of devices increases and should easily allow bidirectional communication between devices and the cloud.

## 10. Data handling

The data that you receive from your IoT hardware needs to be properly stored and analysed. This becomes challenging as more and more devices are introduced into the system and the volume of data increases. While selecting a platform, try to find out how your vendor handles data from your system. Keep in mind that, at times, the pricing model heavily depends on the data. Also, make sure to be well informed about data ownership rights and regulations.

## 11. Security features

While selecting an IoT platform, one must choose a platform with good security solutions. You should also check if the vendor has the latest security certifications and compli-